

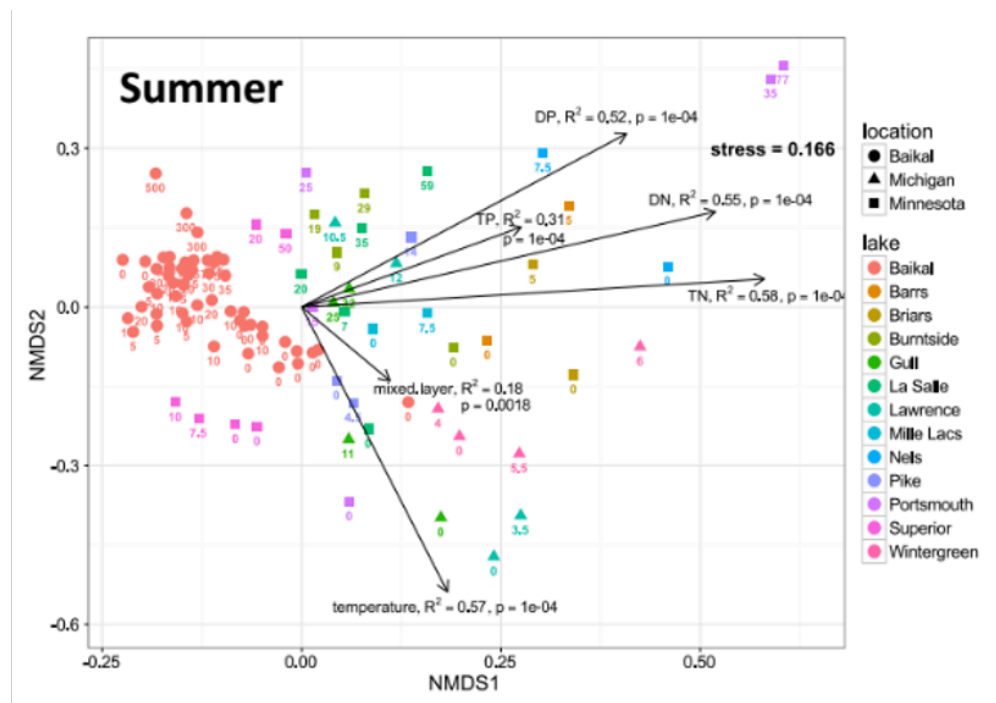
Lecture 17 - Principal Component Analysis (PCA)

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Lecture 16: Review

Review

- Multivariate data
- Multivariate statistics in ecology: overview
- Eigenvectors, eigenvalues, components
- Distance and dissimilarity in MV space
- Data standardization
- Graphics
- Screening MV data
- MANOVA



Review: Eigenvectors and Components

Eigenvectors, eigenvalues and components

- Common goal of MV analysis is variable reduction: can we derive new variables (based on linear combinations of “original” variables) that explain variation in data?
- For data set with $i = 1$ to n objects and $j = 1$ to p original variables we seek new variables (principal components) using the equation:

$$z_{ik} = c_1 y_{i1} + c_2 y_{i2} + \dots + c_j y_{ij} + \dots + c_p y_{ip}$$

Review: Component Interpretation

- z_{ik} is value of new variable k for object i

- y_{i1} - y_{ip} are values of original variables for object i
- c_1 - c_p are coefficients that show importance of the original variables to new derived variable

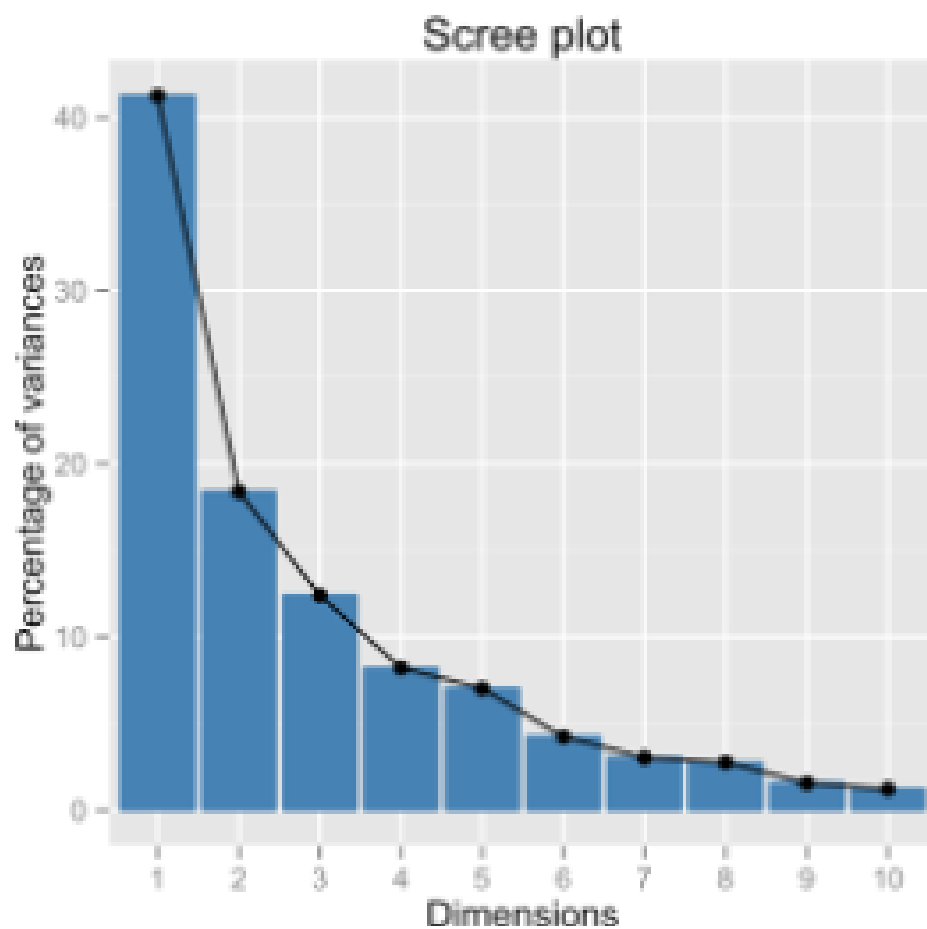
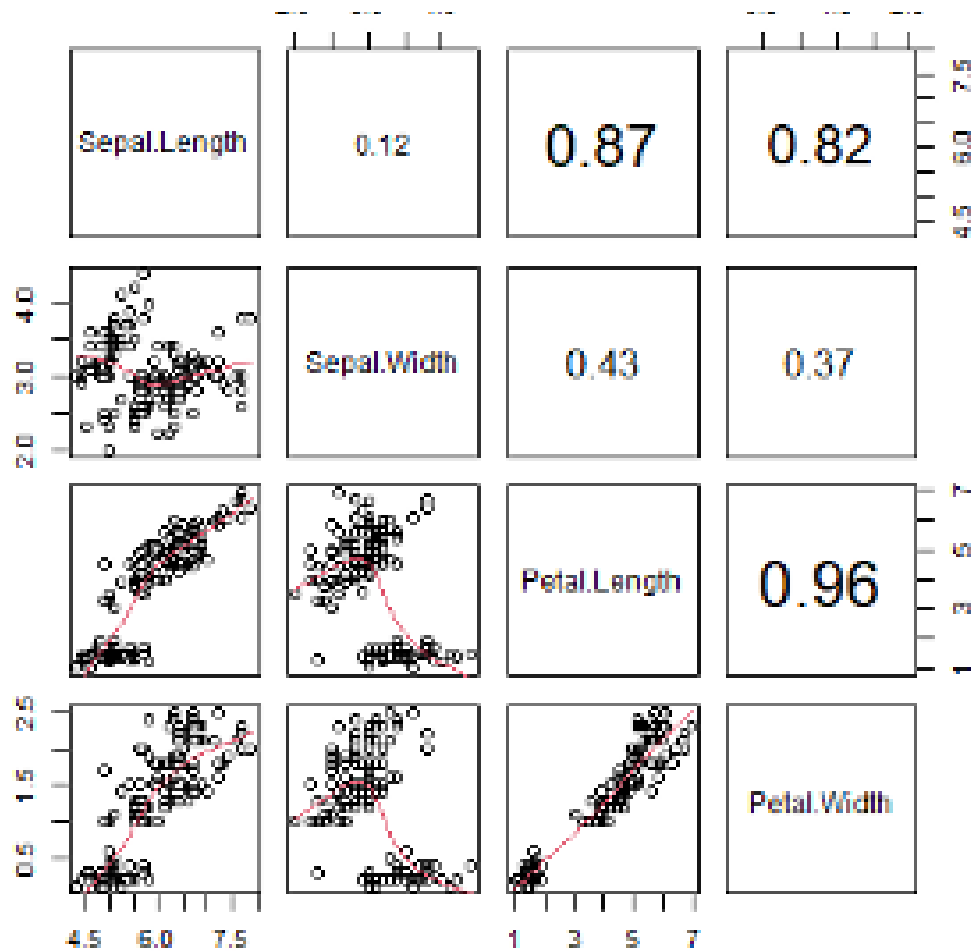
$$z_{ik} = c_1 y_{i1} + c_2 y_{i2} + \dots c_j y_{ij} + \dots + c_p y_{ip}$$

Review: Component Properties

Eigenvectors, eigenvalues and components

Derived variables are found so that:

- First derived variable explains most of the variation in the data
- Second most of the remaining variation
- And so on...
- As many derived variables as original variables (p)
- Derived variables are uncorrelated with each other



Review: Eigenvalues and Eigenvectors

Eigenvectors, eigenvalues and components

- Eigenvalues (latent roots) represent amount of variation in data explained by the new $k=1$ to p derived variables ($\lambda_1, \lambda_2 \dots \lambda_p$).
- Eigenvalues are population parameters and are estimated using ML to get sample statistics ($l_1, l_2 \dots l_p$)
- Eigenvectors are lists of coefficients (c) that show contribution of original variables to new, derived variables
- Each new variable has an eigenvalue and an eigenvector
- New variables (components) are derived from a $p \times p$ covariance or correlation matrix of original variables

Lecture 17: PCA Goals and Introduction

- Common goals of MV data analysis are variable reduction (finding derived variables that summarize data) and exploration of patterns in data (scaling/ordination)
- Can use association (correlation/ covariance) matrices (PCA) or dissimilarity measures (MDS)
- In PCA: take p old variables and transform them into p “new/derived” uncorrelated variables (principal components)

Data for PCA Analysis

```
# Load the iris dataset
iris_df <- iris %>% clean_names() %>% mutate(ind = row_number()) %>%
  mutate(species_ind = paste(species, ind, sep="_"))

# get values only
iris_data <- iris_df %>% select(-species, -ind, -species_ind)

# Keep species for later visualization
iris_species <- iris_df %>% select(species, ind, species_ind)

# pivot to long format for viewing
iris_long_df <- iris_df %>%
  pivot_longer(
    cols = -c(species, ind, species_ind),
    names_to = "variable",
    values_to = "values")

iris_df
```

	sepal_length	sepal_width	petal_length	petal_width	species	ind
1	5.1	3.5	1.4	0.2	setosa	1
2	4.9	3.0	1.4	0.2	setosa	2
3	4.7	3.2	1.3	0.2	setosa	3
4	4.6	3.1	1.5	0.2	setosa	4
5	5.0	3.6	1.4	0.2	setosa	5
6	5.4	3.9	1.7	0.4	setosa	6
7	4.6	3.4	1.4	0.3	setosa	7
8	5.0	3.4	1.5	0.2	setosa	8
9	4.4	2.9	1.4	0.2	setosa	9
10	4.9	3.1	1.5	0.1	setosa	10
11	5.4	3.7	1.5	0.2	setosa	11
12	4.8	3.4	1.6	0.2	setosa	12
13	4.8	3.0	1.4	0.1	setosa	13
14	4.3	3.0	1.1	0.1	setosa	14
15	5.8	4.0	1.2	0.2	setosa	15

16	5.7	4.4	1.5	0.4	setosa	16
17	5.4	3.9	1.3	0.4	setosa	17
18	5.1	3.5	1.4	0.3	setosa	18
19	5.7	3.8	1.7	0.3	setosa	19
20	5.1	3.8	1.5	0.3	setosa	20
21	5.4	3.4	1.7	0.2	setosa	21
22	5.1	3.7	1.5	0.4	setosa	22
23	4.6	3.6	1.0	0.2	setosa	23
24	5.1	3.3	1.7	0.5	setosa	24
25	4.8	3.4	1.9	0.2	setosa	25
26	5.0	3.0	1.6	0.2	setosa	26
27	5.0	3.4	1.6	0.4	setosa	27
28	5.2	3.5	1.5	0.2	setosa	28
29	5.2	3.4	1.4	0.2	setosa	29
30	4.7	3.2	1.6	0.2	setosa	30
31	4.8	3.1	1.6	0.2	setosa	31
32	5.4	3.4	1.5	0.4	setosa	32
33	5.2	4.1	1.5	0.1	setosa	33
34	5.5	4.2	1.4	0.2	setosa	34
35	4.9	3.1	1.5	0.2	setosa	35
36	5.0	3.2	1.2	0.2	setosa	36
37	5.5	3.5	1.3	0.2	setosa	37
38	4.9	3.6	1.4	0.1	setosa	38
39	4.4	3.0	1.3	0.2	setosa	39
40	5.1	3.4	1.5	0.2	setosa	40
41	5.0	3.5	1.3	0.3	setosa	41
42	4.5	2.3	1.3	0.3	setosa	42
43	4.4	3.2	1.3	0.2	setosa	43
44	5.0	3.5	1.6	0.6	setosa	44
45	5.1	3.8	1.9	0.4	setosa	45
46	4.8	3.0	1.4	0.3	setosa	46
47	5.1	3.8	1.6	0.2	setosa	47
48	4.6	3.2	1.4	0.2	setosa	48
49	5.3	3.7	1.5	0.2	setosa	49
50	5.0	3.3	1.4	0.2	setosa	50
51	7.0	3.2	4.7	1.4	versicolor	51
52	6.4	3.2	4.5	1.5	versicolor	52
53	6.9	3.1	4.9	1.5	versicolor	53
54	5.5	2.3	4.0	1.3	versicolor	54
55	6.5	2.8	4.6	1.5	versicolor	55
56	5.7	2.8	4.5	1.3	versicolor	56
57	6.3	3.3	4.7	1.6	versicolor	57
58	4.9	2.4	3.3	1.0	versicolor	58
59	6.6	2.9	4.6	1.3	versicolor	59
60	5.2	2.7	3.9	1.4	versicolor	60
61	5.0	2.0	3.5	1.0	versicolor	61
62	5.9	3.0	4.2	1.5	versicolor	62
63	6.0	2.2	4.0	1.0	versicolor	63
64	6.1	2.9	4.7	1.4	versicolor	64
65	5.6	2.9	3.6	1.3	versicolor	65
66	6.7	3.1	4.4	1.4	versicolor	66
67	5.6	3.0	4.5	1.5	versicolor	67
68	5.8	2.7	4.1	1.0	versicolor	68
69	6.2	2.2	4.5	1.5	versicolor	69
70	5.6	2.5	3.9	1.1	versicolor	70
71	5.9	3.2	4.8	1.8	versicolor	71
72	6.1	2.8	4.0	1.3	versicolor	72

73	6.3	2.5	4.9	1.5 versicolor	73
74	6.1	2.8	4.7	1.2 versicolor	74
75	6.4	2.9	4.3	1.3 versicolor	75
76	6.6	3.0	4.4	1.4 versicolor	76
77	6.8	2.8	4.8	1.4 versicolor	77
78	6.7	3.0	5.0	1.7 versicolor	78
79	6.0	2.9	4.5	1.5 versicolor	79
80	5.7	2.6	3.5	1.0 versicolor	80
81	5.5	2.4	3.8	1.1 versicolor	81
82	5.5	2.4	3.7	1.0 versicolor	82
83	5.8	2.7	3.9	1.2 versicolor	83
84	6.0	2.7	5.1	1.6 versicolor	84
85	5.4	3.0	4.5	1.5 versicolor	85
86	6.0	3.4	4.5	1.6 versicolor	86
87	6.7	3.1	4.7	1.5 versicolor	87
88	6.3	2.3	4.4	1.3 versicolor	88
89	5.6	3.0	4.1	1.3 versicolor	89
90	5.5	2.5	4.0	1.3 versicolor	90
91	5.5	2.6	4.4	1.2 versicolor	91
92	6.1	3.0	4.6	1.4 versicolor	92
93	5.8	2.6	4.0	1.2 versicolor	93
94	5.0	2.3	3.3	1.0 versicolor	94
95	5.6	2.7	4.2	1.3 versicolor	95
96	5.7	3.0	4.2	1.2 versicolor	96
97	5.7	2.9	4.2	1.3 versicolor	97
98	6.2	2.9	4.3	1.3 versicolor	98
99	5.1	2.5	3.0	1.1 versicolor	99
100	5.7	2.8	4.1	1.3 versicolor	100
101	6.3	3.3	6.0	2.5 virginica	101
102	5.8	2.7	5.1	1.9 virginica	102
103	7.1	3.0	5.9	2.1 virginica	103
104	6.3	2.9	5.6	1.8 virginica	104
105	6.5	3.0	5.8	2.2 virginica	105
106	7.6	3.0	6.6	2.1 virginica	106
107	4.9	2.5	4.5	1.7 virginica	107
108	7.3	2.9	6.3	1.8 virginica	108
109	6.7	2.5	5.8	1.8 virginica	109
110	7.2	3.6	6.1	2.5 virginica	110
111	6.5	3.2	5.1	2.0 virginica	111
112	6.4	2.7	5.3	1.9 virginica	112
113	6.8	3.0	5.5	2.1 virginica	113
114	5.7	2.5	5.0	2.0 virginica	114
115	5.8	2.8	5.1	2.4 virginica	115
116	6.4	3.2	5.3	2.3 virginica	116
117	6.5	3.0	5.5	1.8 virginica	117
118	7.7	3.8	6.7	2.2 virginica	118
119	7.7	2.6	6.9	2.3 virginica	119
120	6.0	2.2	5.0	1.5 virginica	120
121	6.9	3.2	5.7	2.3 virginica	121
122	5.6	2.8	4.9	2.0 virginica	122
123	7.7	2.8	6.7	2.0 virginica	123
124	6.3	2.7	4.9	1.8 virginica	124
125	6.7	3.3	5.7	2.1 virginica	125
126	7.2	3.2	6.0	1.8 virginica	126
127	6.2	2.8	4.8	1.8 virginica	127
128	6.1	3.0	4.9	1.8 virginica	128
129	6.4	2.8	5.6	2.1 virginica	129

130	7.2	3.0	5.8	1.6	virginica	130
131	7.4	2.8	6.1	1.9	virginica	131
132	7.9	3.8	6.4	2.0	virginica	132
133	6.4	2.8	5.6	2.2	virginica	133
134	6.3	2.8	5.1	1.5	virginica	134
135	6.1	2.6	5.6	1.4	virginica	135
136	7.7	3.0	6.1	2.3	virginica	136
137	6.3	3.4	5.6	2.4	virginica	137
138	6.4	3.1	5.5	1.8	virginica	138
139	6.0	3.0	4.8	1.8	virginica	139
140	6.9	3.1	5.4	2.1	virginica	140
141	6.7	3.1	5.6	2.4	virginica	141
142	6.9	3.1	5.1	2.3	virginica	142
143	5.8	2.7	5.1	1.9	virginica	143
144	6.8	3.2	5.9	2.3	virginica	144
145	6.7	3.3	5.7	2.5	virginica	145
146	6.7	3.0	5.2	2.3	virginica	146
147	6.3	2.5	5.0	1.9	virginica	147
148	6.5	3.0	5.2	2.0	virginica	148
149	6.2	3.4	5.4	2.3	virginica	149
150	5.9	3.0	5.1	1.8	virginica	150

species_ind

1	setosa_1
2	setosa_2
3	setosa_3
4	setosa_4
5	setosa_5
6	setosa_6
7	setosa_7
8	setosa_8
9	setosa_9
10	setosa_10
11	setosa_11
12	setosa_12
13	setosa_13
14	setosa_14
15	setosa_15
16	setosa_16
17	setosa_17
18	setosa_18
19	setosa_19
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21	setosa_21
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31	setosa_31
32	setosa_32
33	setosa_33
34	setosa_34
35	setosa_35

36	setosa_36
37	setosa_37
38	setosa_38
39	setosa_39
40	setosa_40
41	setosa_41
42	setosa_42
43	setosa_43
44	setosa_44
45	setosa_45
46	setosa_46
47	setosa_47
48	setosa_48
49	setosa_49
50	setosa_50
51	versicolor_51
52	versicolor_52
53	versicolor_53
54	versicolor_54
55	versicolor_55
56	versicolor_56
57	versicolor_57
58	versicolor_58
59	versicolor_59
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62	versicolor_62
63	versicolor_63
64	versicolor_64
65	versicolor_65
66	versicolor_66
67	versicolor_67
68	versicolor_68
69	versicolor_69
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146 virginica_146
147 virginica_147
148 virginica_148

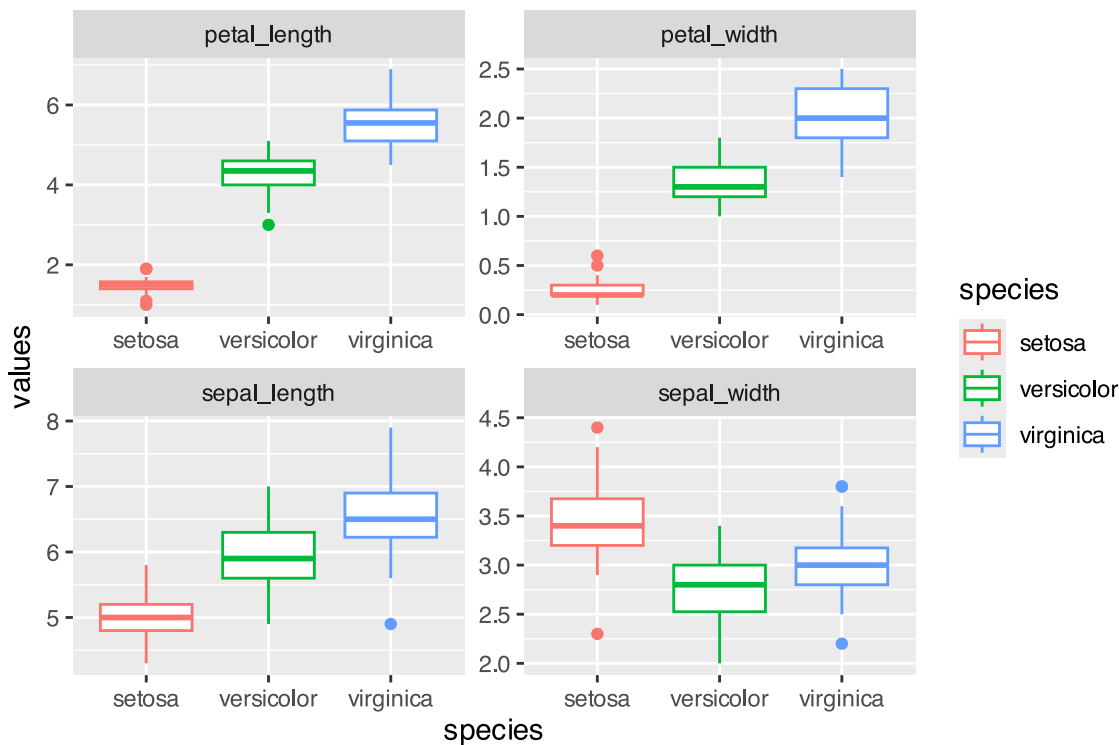
```
149 virginica_149
150 virginica_150
```

Step 1: Explore the Iris Dataset

As in every case you should be looking at the data first - every time...

Right is the data on iris from a long dataframe

```
overview_plot <- iris_long_df %>%
  ggplot(aes(species, values, color=species)) +
  geom_boxplot() +
  facet_wrap(~variable, scales = "free")
overview_plot
```



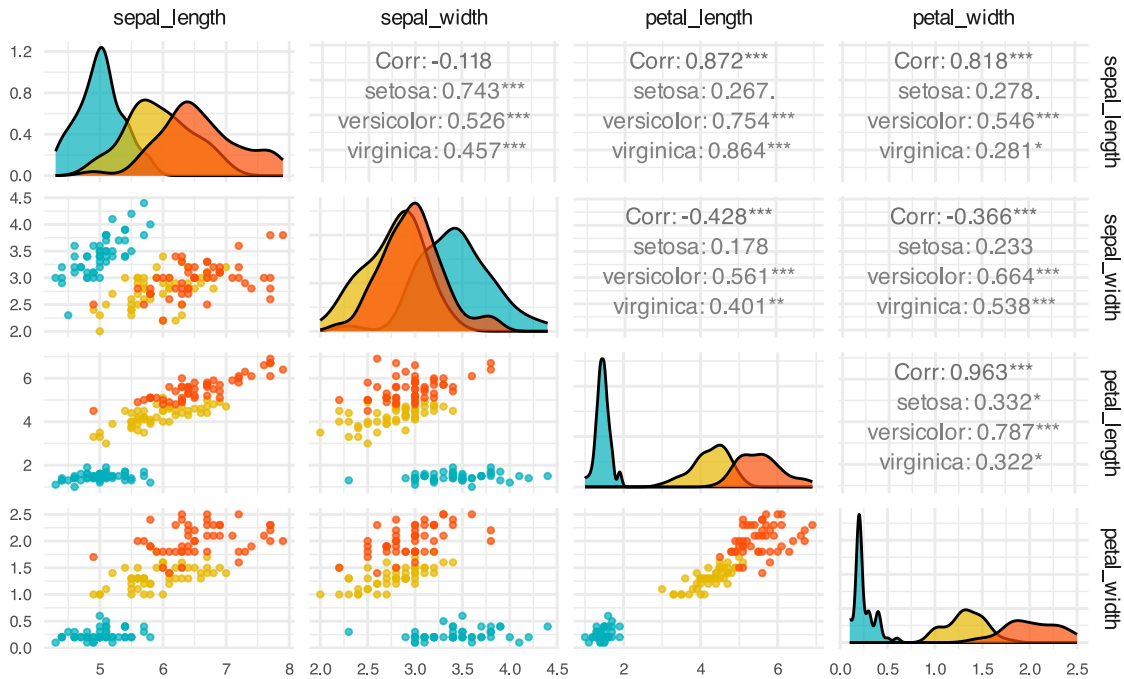
What is PCA? Goals and Overview

Principal Component Analysis Goals:

- **Variable Reduction:** Transform many correlated variables into fewer uncorrelated components
- **Data Exploration:** Visualize patterns and relationships in high-dimensional data
- **Noise Reduction:** Focus on the most important sources of variation
- **Dimension Reduction:** Make complex datasets easier to analyze and interpret

Today's Example: Iris flower measurements - can we reduce 4 measurements to 2-3 components that capture most variation?

The Challenge: 4 Dimensions of Iris Data



High-Dimensional Data Visualization

Principal Component Analysis Goals:

- **Variable Reduction:** Transform many correlated variables into fewer uncorrelated components
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- **Dimension Reduction:** Make complex datasets easier to analyze and interpret

Today's Example: Iris flower measurements - can we reduce 4 measurements to fewer components that capture most variation?

PCA Assumptions - Critical to Check First!

Key Assumptions:

1. **Linear relationships** between variables
2. **No extreme outliers** (can distort results)
3. **Variables should be correlated** (if not, PCA won't reduce dimensions)
4. **Adequate sample size** (generally $n > 50$, preferably $n > 100$)
5. **No missing data** (complete cases only)
6. **Consider standardization** when variables have different scales

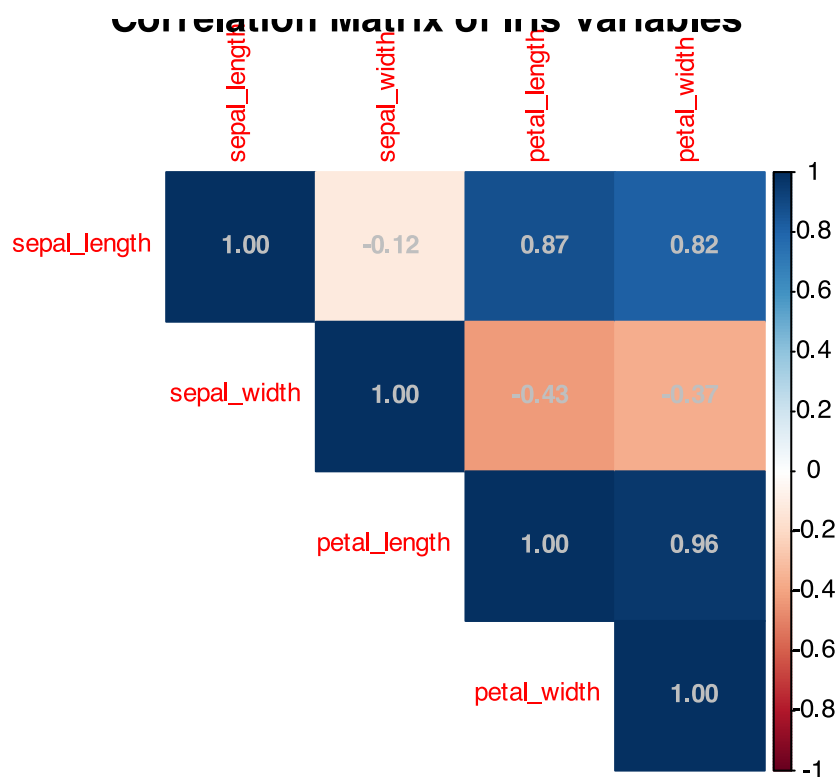
Important: PCA works best when original variables are moderately to highly correlated!

Let's check these assumptions with our iris data...

Step 2: Check PCA Assumptions - Correlations

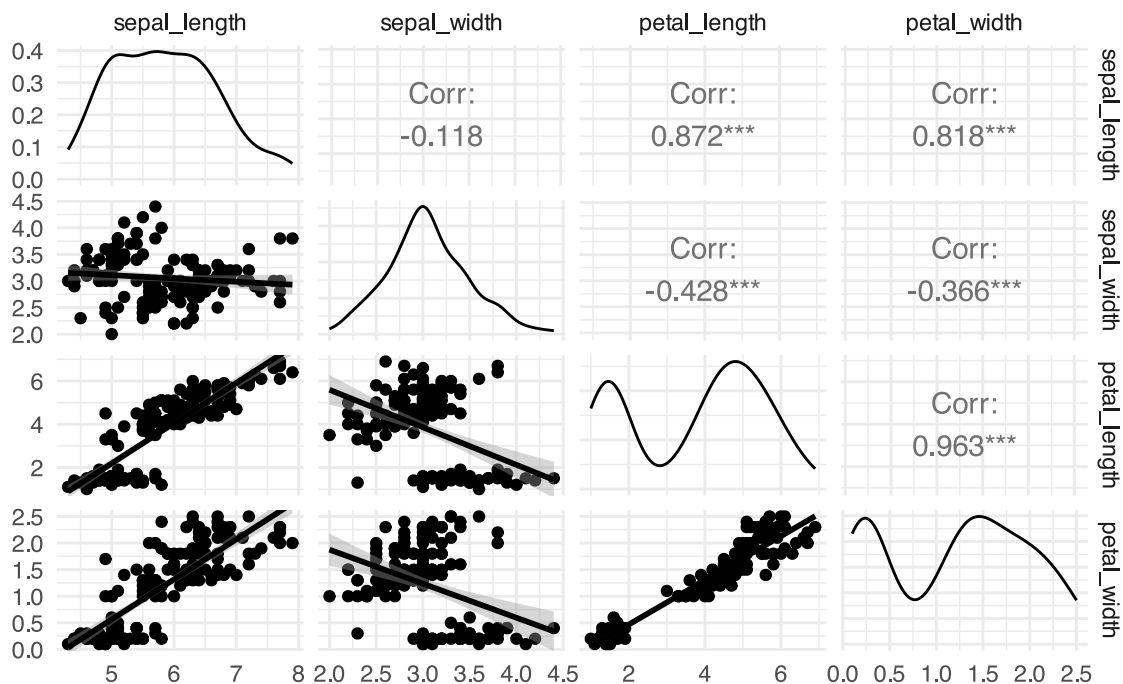
```
[1] "Correlation Matrix:"
```

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000	-0.118	0.872	0.818
sepal_width	-0.118	1.000	-0.428	-0.366
petal_length	0.872	-0.428	1.000	0.963
petal_width	0.818	-0.366	0.963	1.000



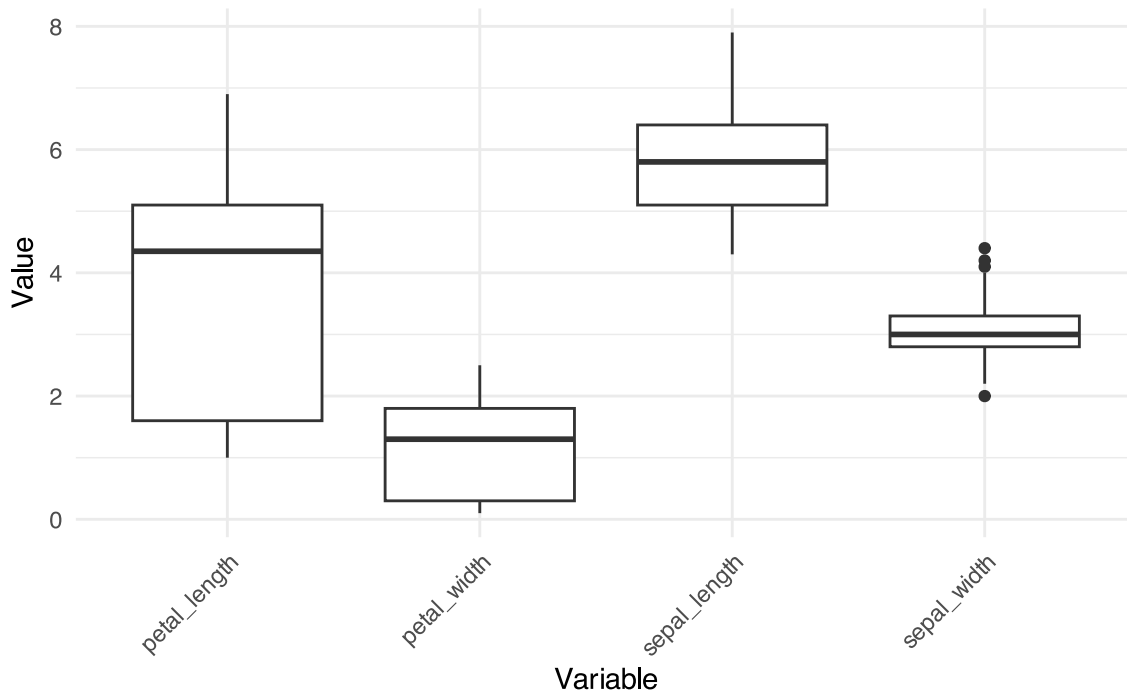
Step 2: Check PCA Assumptions - Linearity

Pairwise Relationships in Iris Data



Step 2: Check PCA Assumptions - Outliers

Check for Outliers in Iris Variables



Step 3: Standardize the Data

STANDARDIZATION: Making all variables comparable

Why standardize?

Our measurements have different units and scales:

- Sepal length: ranges from ~4-8 cm
- Sepal width: ranges from ~2-4 cm
- Petal length: ranges from ~1-7 cm
- Petal width: ranges from ~0.1-2.5 cm

Without standardization, PCA would be dominated by variables with larger numbers (like petal length) simply because they have bigger values, not because they're more important biologically

What does standardization do?

- Converts each variable to have:
 - Mean = 0 (centered at zero)
 - Standard deviation = 1 (same spread)
 - This gives all variables equal weight in the analysis

How to interpret standardized values: Example: A sepal length of 5.1 cm might become -0.9 after standardization, meaning it's 0.9 standard deviations below the average sepal length

```
[1] "Means after standardization (should be ~0):"
```

```
sepal_length  sepal_width  petal_length  petal_width
           0           0           0           0
```

```
[1] "Standard deviations after standardization (should be 1):"
```

```
sepal_length  sepal_width petal_length  petal_width
            1            1            1            1
```

Step 4: Perform PCA - The Mathematics

What is PCA doing?

Principal Component Analysis finds new variables (called components) that capture the most variation in your data. Think of it as finding the “best viewing angles” to see differences between flowers.

The mathematics (simplified):

- PCA rotates your data to find the direction with maximum spread (PC1)
- Then finds the next direction with maximum spread perpendicular to PC1 (PC2)
- Continues until it has as many components as original variables (4 in our case)

Why center = FALSE and scale = FALSE?

We already standardized our data in Step 3, so we tell R not to do it again: - center = FALSE: Don't subtract the mean (we already did) - scale = FALSE: Don't divide by standard deviation (we already did)

What the summary shows:

- **Standard deviation:** How much variation each component captures
- **Proportion of Variance:** Percentage of total variation explained by each component
- **Cumulative Proportion:** Running total of variance explained

```
# Perform PCA on standardized data
iris_pca <- prcomp(iris_scaled, center = FALSE, scale. = FALSE)
# Note: center and scale are FALSE because we already standardized

# Alternative using vegan package
iris_pca_vegan <- rda(iris_scaled)

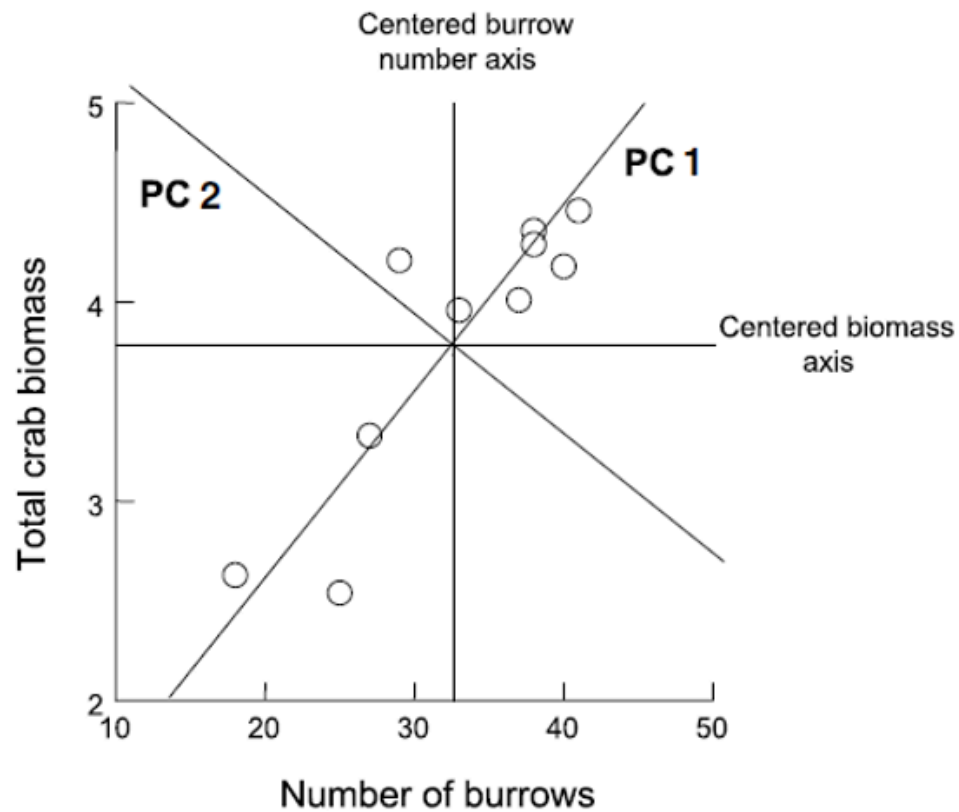
# Summary of PCA results
summary(iris_pca)
```

```
Importance of components:

            PC1      PC2      PC3      PC4
Standard deviation   1.7084 0.9560 0.38309 0.14393
Proportion of Variance 0.7296 0.2285 0.03669 0.00518
Cumulative Proportion 0.7296 0.9581 0.99482 1.00000
```

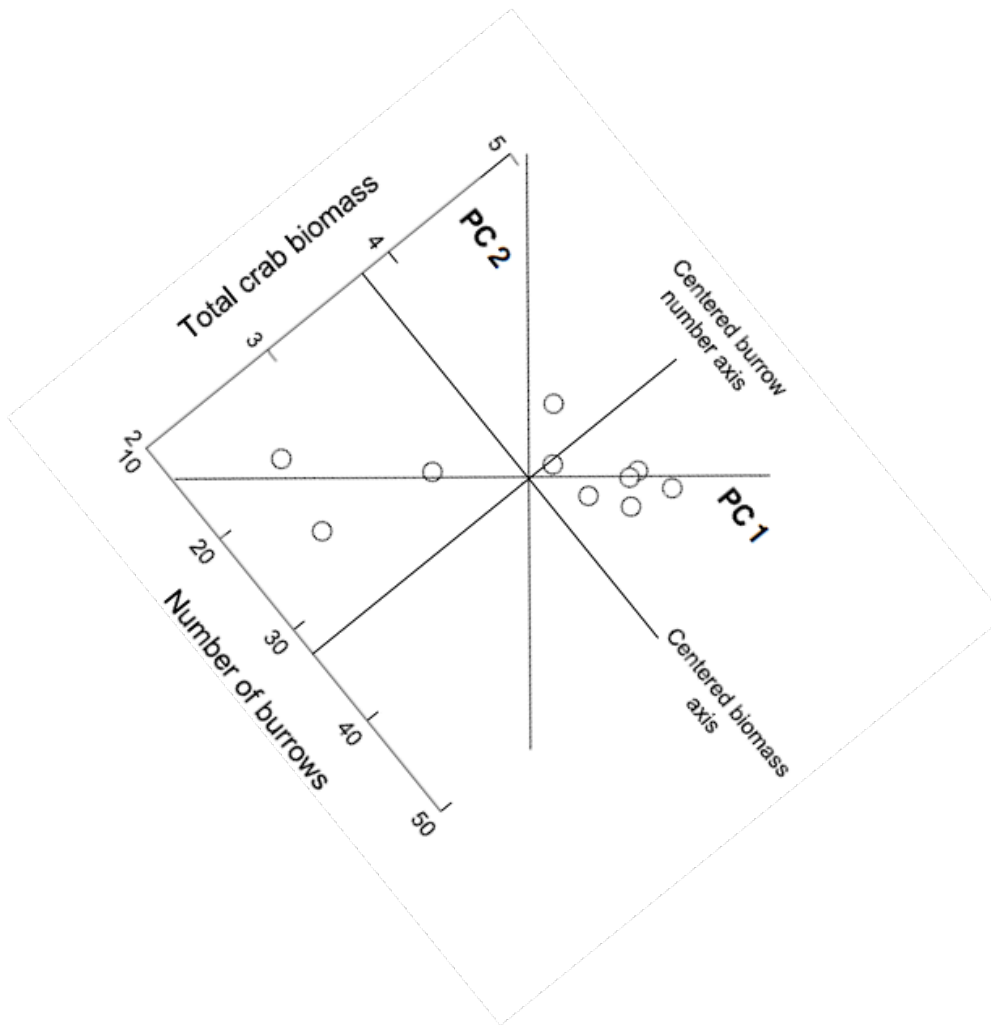
Deriving Components: 2D Visualization

How are new uncorrelated components derived? One way to think it is in terms of axis rotation Consider a 2-variable dataset:



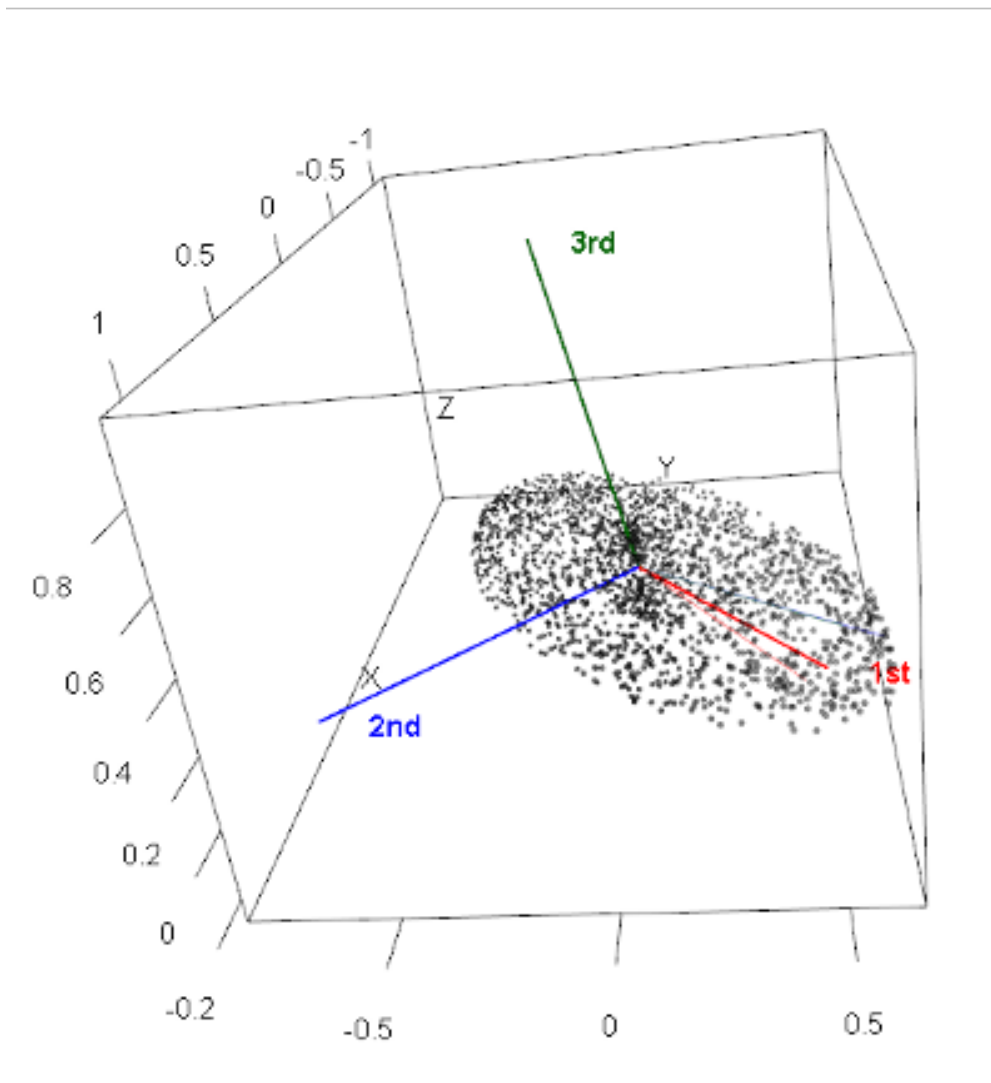
Component Derivation: Axis Rotation

Goal is to “rotate the axes” around center of the data “cloud” in such a way that most of the variation lies along the first axis. Then find second axis that explains the second-most variation AND is orthogonal to first axis.



Component Derivation: Multivariate Extension

Easy to picture in 2D (or even 3D), but harder in multivariate space Practically, components are “extracted” from a covariance or correlation matrix among original variables Will extract as many principal components as original variables



Component Information: Eigenvalues and Eigenvectors

- Get two important pieces of information from PCA: eigenvectors and eigenvalues
- Eigenvalues (latent roots)- how much of the variation is explained by each component?
- Eigenvectors- list of coefficients for original variables. There are p coefficients in an eigenvector and p eigenvectors
- Correlation bw original variables will result in fewer components explaining more variance; variable reduction will fail if original variables are not correlated

Step 4: Understanding Eigenvalues and Variance

Understanding Eigenvalues and Variance

What are eigenvalues?

- Eigenvalues tell us how much variation each principal component captures.
- Larger eigenvalues = more important components.

Key terms explained:

- **Eigenvalue:** The amount of variance captured by each component (always positive)
- **Proportion of Variance:** What percentage of total variation this component explains
- **Cumulative Variance:** Running total - helps us decide how many components we need

How to read the results:

- If PC1 has eigenvalue = 2.9, it captures 2.9 “units” of variance
- If Prop_Variance = 0.728, PC1 explains 72.8% of all variation in the data
- If Cumsum_Variance = 0.959 at PC2, the first 2 components together explain 95.9% of variation

Why this matters:

This table helps us decide how many components to keep.

- If 2 components explain 95% of variance, we’ve successfully reduced 4 variables to 2
- We only lose 5% of information without including the other variables!

```
# Extract eigenvalues (variance explained by each component)
eigenvalues <- iris_pca$sdev^2
prop_variance <- eigenvalues / sum(eigenvalues)
cumsum_variance <- cumsum(prop_variance)

# Create a summary table
pca_summary <- data.frame(
  Component = paste0("PC", 1:length(eigenvalues)),
  Eigenvalue = eigenvalues,
  Prop_Variance = prop_variance,
  Cumsum_Variance = cumsum_variance
)

print("PCA Summary:")
```

```
[1] "PCA Summary:"
```

```
kable(pca_summary, digits = 3)
```

Component	Eigenvalue	Prop_Variance	Cumsum_Variance
PC1	2.918	0.730	0.730
PC2	0.914	0.229	0.958
PC3	0.147	0.037	0.995
PC4	0.021	0.005	1.000

Step 5: Determine Number of Components - Scree Plot

What is a Scree Plot?

A scree plot shows how much variance each component explains, helping us decide how many components we need. The name comes from the geological term “scree” - loose rocks at the base of a cliff - because the plot often looks like a steep cliff followed by rubble.

How to read a Scree Plot:

- **Y-axis:** Percentage of variance explained by each component
- **X-axis:** Component number (PC1, PC2, etc.)
- **The pattern:** Usually shows a steep drop followed by a leveling off

The “Elbow Method”:

Look for where the line “bends” or forms an elbow:

- Components before the elbow = important (steep slope)

- Components after the elbow = less important (gentle slope)
- Keep components up to and including the elbow

What to look for in our plot:

- If PC1 explains 70% and PC2 explains 20%, but PC3 only explains 5%, the elbow is at PC2
- This suggests keeping the first 2 components
- The dramatic drop from PC1 to PC2, then gentle decline after, confirms our dimension reduction worked well

Step 5: Component Selection Rules

```
# Eigenvalue > 1 rule (Kaiser criterion)
components_to_keep <- sum(eigenvalues > 1)
print(paste("Components with eigenvalue > 1:", components_to_keep))
```

```
[1] "Components with eigenvalue > 1: 1"
```

```
# Components explaining at least 80% of variance
components_80_percent <- which(cumsum_variance >= 0.80)[1]
print(paste("Components needed for 80% variance:", components_80_percent))
```

```
[1] "Components needed for 80% variance: 2"
```

Step 6: Interpret the Components - Loadings

What are Component Loadings

Loadings tell us how much each original variable contributes to each principal component. Think of them as “recipes” that show how to mix your original measurements to create the new components.

How to read the loadings table

- **Values range from -1 to +1** (like correlations)
- **Large positive values** (e.g., 0.8): This variable contributes strongly in the positive direction
- **Large negative values** (e.g., -0.8): This variable contributes strongly in the negative direction
- **Values near 0**: This variable doesn’t contribute much to this component

Interpreting the patterns:

- **If all loadings have similar signs**: Component represents overall size (all measurements increase/decrease together)
- **If loadings have mixed signs**: Component represents shape or proportions (some measurements increase while others decrease)
- **Dominant variables**: Variables with the largest absolute loadings drive that component’s meaning

Example interpretation:

If PC1 has all negative loadings around -0.5, it means:

- Flowers with high PC1 scores have small values for ALL measurements
- This component captures “overall flower size”
- The negative sign just indicates direction (could flip signs and interpretation)

```
# Component loadings (how much each original variable contributes)
loadings_df <- data.frame(
```

```
Variable = rownames(iris_pca$rotation),
PC1 = iris_pca$rotation[, 1],
PC2 = iris_pca$rotation[, 2],
PC3 = iris_pca$rotation[, 3],
PC4 = iris_pca$rotation[, 4]
)

print("Component Loadings:")
```

```
[1] "Component Loadings:"
```

```
loadings_df
```

	Variable	PC1	PC2	PC3	PC4
sepal_length	sepal_length	0.5210659	-0.37741762	0.7195664	0.2612863
sepal_width	sepal_width	-0.2693474	-0.92329566	-0.2443818	-0.1235096
petal_length	petal_length	0.5804131	-0.02449161	-0.1421264	-0.8014492
petal_width	petal_width	0.5648565	-0.06694199	-0.6342727	0.5235971

Step 6: Eigenvector Properties

Key properties of eigenvectors/loadings:

- **Unit length:** Each eigenvector has length 1 (sum of squares = 1)
- **Orthogonal:** Eigenvectors are perpendicular to each other (dot product = 0)
- **Ordered by importance:** First eigenvector (PC1) explains most variance

The complete picture:

- **Eigenvectors** = The directions (loadings)
- **Eigenvalues** = The importance of each direction (variance explained)
- Together they fully describe the PCA transformation

Step 6b: Visualization of Component Loadings

What does this plot show?

This is a visual representation of the loadings table, showing how each original variable contributes to PC1 and PC2. It's like a map of how your original measurements relate to the new principal components.

How to read the plot:

- Arrows represent your original variables (sepal_length, sepal_width, etc.)
- Arrow direction shows which PC the variable contributes to
- Arrow length indicates the strength of contribution (longer = stronger)
- Arrow color shows the overall contribution magnitude (red = highest, blue = lowest)
- The circle represents the maximum possible contribution

Key interpretations from this plot:

- PC1 (horizontal axis, 73% variance):
 - All arrows point roughly left (negative direction)
 - All variables contribute almost equally to PC1
 - This confirms PC1 represents "overall flower size"

PC2 (vertical axis, 22.9% variance):

- Sepal_width points down (negative)
- Other variables point slightly up (positive)
- This creates a contrast: sepal width vs. everything else
- PC2 captures “flower shape” - wide sepals vs. long petals

Loading Plot Interpretation

What the arrow positions tell us:

- Variables pointing in same direction = positively correlated
- Variables at 90° angles = uncorrelated
- Variables pointing opposite directions = negatively correlated

The practical meaning:

- Flowers with high PC1 scores have large values for all measurements
- Flowers with high PC2 scores have narrow sepals but long/wide petals
- The plot confirms our dimension reduction worked - we’ve captured 95.9% of variation in just 2 dimensions!

Step 7: PCA Biplot - The Main Result

Key insights from this biplot:

Species separation:

- **Setosa (blue):** Clearly separated on the left (negative PC1)
- **Versicolor (yellow):** In the middle
- **Virginica (red):** On the right (positive PC1)
- PCA successfully separates species without being told about them!

Understanding flower characteristics:

- **Setosa flowers:** Small overall (negative PC1), relatively wide sepals (positive PC2)
- **Virginica flowers:** Large overall (positive PC1), especially long petals
- **Versicolor flowers:** Intermediate in most characteristics

Variable relationships:

- Petal measurements point together → highly correlated
- Sepal width points differently → captures different information
- All arrows point right → all measurements increase from setosa to virginica

Step 7: PCA Scores Plot - Alternative Visualization

```
# Alternative scores plot
ggplot(pca_scores, aes(x = PC1, y = PC2, color = species)) +
  geom_point(size = 3, alpha = 0.7) +
  stat_ellipse(level = 0.68, linetype = 2) + # Add ellipses
  labs(title = "PCA Scores Plot",
        subtitle = paste0("PC1 explains ", round(prop_variance[1]*100, 1),
                           "% of variance, PC2 explains ", round(prop_variance[2]*100, 1), "%"),
        x = paste0("PC1 (", round(prop_variance[1]*100, 1), "%)"),
        y = paste0("PC2 (", round(prop_variance[2]*100, 1), "%)"),
        color = "Species") +
  theme_minimal() +
  scale_color_manual(values = c("#00AFBB", "#E7B800", "#FC4E07"))
```

Step 8: Interpret PC1 Results

Understanding PC1 Loadings:

The loadings show how each original variable contributes to PC1:

- **Sepal length: 0.521** - Strong positive contribution
- **Sepal width: -0.269** - Moderate negative contribution
- **Petal length: 0.580** - Strong positive contribution
- **Petal width: 0.565** - Strong positive contribution

```
# What does PC1 represent?  
pc1_loadings <- iris_pca$rotation[, 1]  
print("PC1 Loadings (all variables contribute similarly):")
```

```
[1] "PC1 Loadings (all variables contribute similarly):"
```

```
round(pc1_loadings, 3)
```

sepal_length	sepal_width	petal_length	petal_width
0.521	-0.269	0.580	0.565

```
cat("\nPC1 Interpretation: Overall flower size")
```

```
PC1 Interpretation: Overall flower size
```

```
cat("\n- All variables have similar negative loadings")
```

```
- All variables have similar negative loadings
```

```
cat("\n- Higher PC1 values = smaller flowers overall")
```

```
- Higher PC1 values = smaller flowers overall
```

```
cat("\n- Lower PC1 values = larger flowers overall")
```

```
- Lower PC1 values = larger flowers overall
```

PC1 Interpretation: Overall Flower Size

PC1 Interpretation: Overall flower size (with a twist)

Note: The output says “all variables have similar negative loadings” but the actual values show mostly positive loadings. This is likely due to a sign flip - PCA signs can be arbitrary. Let’s interpret based on the actual values shown:

- **Three variables (sepal length, petal length, petal width) have similar positive loadings** (~0.52-0.58)
- **Sepal width has a negative loading** (-0.269)
- This means PC1 captures flowers where length and width measurements (except sepal width) vary together

What PC1 scores mean:

- **Higher PC1 values** = Longer petals, longer sepals, wider petals, but narrower sepals
- **Lower PC1 values** = Shorter petals, shorter sepals, narrower petals, but wider sepals
- PC1 essentially captures “overall flower size except sepal width goes opposite”

Biological interpretation:

PC1 distinguishes between:

- Small flowers with relatively wide sepals (negative PC1) - typical of setosa
- Large flowers with relatively narrow sepals (positive PC1) - typical of virginica

Step 8: Interpret PC2 Results

Understanding PC2 Loadings:

The loadings show how each original variable contributes to PC2:

- **Sepal length: -0.377** - Moderate negative contribution
- **Sepal width: -0.923** - Very strong negative contribution
- **Petal length: -0.024** - Almost no contribution
- **Petal width: -0.067** - Very small negative contribution

```
# What does PC2 represent?
pc2_loadings <- iris_pca$rotation[, 2]
print("PC2 Loadings:")
```

```
[1] "PC2 Loadings:"
```

```
round(pc2_loadings, 3)
```

```
sepal_length  sepal_width  petal_length  petal_width
      -0.377       -0.923       -0.024       -0.067
```

```
cat("\nPC2 Interpretation: Flower shape contrast")
```

```
PC2 Interpretation: Flower shape contrast
```

```
cat("\n- Positive loadings: sepal width")
```

- Positive loadings: sepal width

```
cat("\n- Negative loadings: petal length and width, sepal length")
```

- Negative loadings: petal length and width, sepal length

```
cat("\n- Higher PC2 = wider sepals relative to petal size")
```

- Higher PC2 = wider sepals relative to petal size

```
cat("\n- Lower PC2 = longer/wider petals relative to sepal width")
```

- Lower PC2 = longer/wider petals relative to sepal width

PC2 Interpretation: Flower Shape Contrast

PC2 Interpretation: Correcting the output

Note: The output says “Positive loadings: sepal width” but the actual value is -0.923 (negative). All loadings are actually negative, with sepal width being the most strongly negative.

What PC2 actually represents:

- All variables have negative loadings, but sepal width is dominant (-0.923)
- Petal measurements contribute very little (-0.024 and -0.067)
- This component is primarily driven by sepal width, with some contribution from sepal length

What PC2 scores mean:

- Higher PC2 values = Smaller measurements overall, especially narrow sepals
- Lower PC2 values = Larger measurements overall, especially wide sepals
- Since sepal width has the strongest loading, PC2 primarily captures sepal width variation

Biological interpretation:

PC2 helps distinguish:

- Flowers with narrow sepals and smaller overall size (positive PC2)
- Flowers with wide sepals and larger overall size (negative PC2)
- This dimension helps separate species that have similar PC1 scores but different sepal proportions

Step 9: How Well Does PCA Work?

```
# Calculate total variance explained by first 2 components
variance_explained_2pc <- sum(prop_variance[1:2])
caption_pca <- paste("Variance explained by first 2 components:", round(variance_explained_2pc
* 100, 1), "%")

# This means we reduced 4 variables to 2 components while retaining most information!
```



```
# Create a summary plot showing dimension reduction success
tibble(
  Component = factor(paste0("PC", 1:4), levels = paste0("PC", 1:4)),
  Variance = prop_variance * 100,
  Cumulative = cumsum_variance * 100
) %>%
  ggplot(aes(x = Component)) +
  geom_col(aes(y = Variance), fill = "lightblue", alpha = 0.7) +
  geom_line(aes(y = Cumulative, group = 1), color = "red", size = 1) +
  geom_point(aes(y = Cumulative), color = "red", size = 3) +
  labs(title = "PCA Dimension Reduction Success",
       subtitle = "Blue bars = individual variance, Red line = cumulative variance",
       caption = caption_pca,
       x = "Principal Component",
       y = "Percentage of Variance Explained") +
  theme_minimal()
```

Summary: What We Learned

Key Findings:

1. **Successful dimension reduction:** 4 variables → 2 components explaining ~96% of variance
2. **PC1 (72.8% variance):** Overall flower size
 - All measurements contribute similarly
 - Separates large from small flowers
3. **PC2 (23.1% variance):** Shape contrast
 - Sepal width vs. petal dimensions
 - Separates flower shape types
4. **Species separation:** PCA naturally groups the three iris species based on their morphological differences

PCA Success Criteria Met:

- ✓ Variables were correlated
- ✓ Linear relationships
- ✓ No major outliers
- ✓ Adequate sample size
- ✓ Clear dimension reduction
- ✓ Interpretable components

When to Use PCA vs. Other Methods

Use PCA when:

- Variables are **continuous and correlated**
- Goal is **dimension reduction** or **data exploration**
- Linear relationships between variables
- Want to **remove redundancy** in measurements

Consider alternatives when:

- Variables are categorical → use MCA (Multiple Correspondence Analysis)
- Focus on **species composition** → use ordination methods like NMDS
- Want to **classify/predict** → use discriminant analysis or machine learning

PCA is excellent for exploring patterns in biological measurements like morphology, physiology, or environmental variables!